AMRITA VISHWA VIDYAPEETHAM
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**Amrita School of AI**
COIMBATORE CAMPUS

LABS · EXERCISES · ASSIGNMENTS

# Classroom 50: Every Graded Item

One page with the accept link, the marks and the deadline for everything that is graded by tests this semester, and the six commands you need to work an item from accept to report.

Prepared by **Dr. Abhijith Anandakrishnan**
Assistant Professor, Amrita School of AI

SECTIONS
AIM A (aim-a-hpc) and AIM B (aim-b-hpc)

ORGANISATION
Amrita-School-of-AI

TOOL
gh student

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OFFERING
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# Read this first

Every lab, coding exercise and assignment in this course is a **Classroom 50** item: a private GitHub repository created for you the moment you accept it, with an autograder that reports on every submission within a couple of minutes. There is nothing to email and nothing to upload. You accept, you work, you submit, you read the report, you submit again if you want to.

Use the links for **your own section**. Accepting an item in the other section creates a repository nobody will grade.

**PITFALL · YOUR GITHUB USERNAME MUST BE THE ONE ON THE ROSTER**

You accepted an invitation to the `Amrita-School-of-AI` organisation at your college address. Only that account can accept items. If `gh student whoami` prints a different name, log out and log in with the right account before accepting anything.

# Part 1: One-time setup

Do this once, on the machine you will work from (your laptop or the cluster login node). It takes about fifteen minutes if your SSH key already works; see `student-guide.pdf` Part 2 if it does not.

```
# 1. GitHub CLI (Debian/Ubuntu; see cli.github.com for other systems)
sudo apt install gh

# 2. Log in, then add the student tool and log in with the scopes it needs
gh auth login                       # choose GitHub.com, SSH, browser login
gh extension install foundation50/gh-student
gh student login
gh student whoami                   # must print your roster username

# 3. Compiler, OpenMP and MPI for local checks
sudo apt install build-essential openmpi-bin libopenmpi-dev libopenblas-dev
```

On the cluster, `gh` and the compilers are already installed; run only steps 2.

# Part 2: Working an item, start to finish

Six commands. The example is Exercise 06 in section A; substitute your section and the item you are working on.

```
# 1. Accept: creates Amrita-School-of-AI/aim-a-hpc-ex06-openmp-reduction-<you>
gh student accept Amrita-School-of-AI aim-a-hpc ex06-openmp-reduction
#    or open the accept link from the tables below in a browser and click Accept

# 2. Clone it (the https:// form asks for a password and fails; use one of these)
gh repo clone Amrita-School-of-AI/aim-a-hpc-ex06-openmp-reduction-<you>
cd aim-a-hpc-ex06-openmp-reduction-<you>

# 3. Read README.md. It is the whole statement, including the marks table.

# 4. Work, and run the public tests as often as you like
make check

# 5. Commit and submit
git add -A
git commit -m "say what you changed"
gh student submit

# 6. Read the report, fix, repeat. Before the second submit:
git pull --rebase origin main
```

## Where the report is

About a minute after `gh student submit`:

- the commit on GitHub gets a **green tick or a red cross**;
- a **Release** appears on your repository, named `submit/<time>-<sha>`. Its text is the grader's report: one line per check, PASS or FAIL, the marks for it, and the reason for every failure;
- the same text is under the run's **Summary** in the Actions tab;
- on classroom50.org, **My submission** shows the score and links to the release.

If there is no release after two minutes, open the Actions tab. A red run whose log ends in a grader error, rather than a failing test, is a problem on my side: send me the link.

## What you may and may not edit

The README of each item names the files that are yours: usually `src/*.c`, sometimes `answers.yaml` or a CSV under `results/`. Everything else is fixed. `.gitignore` and `.github/` are restored on every submit; `tests/` holds copies of the public tests and the grader uses its own copy, so changing them changes nothing. The reference functions marked *given* in a header must stay as they are: the tests compare your versions against them.

## How marks are earned

| KIND OF CHECK | WHAT PASSES IT |
|---|---|
| Compiles | `-Wall -Wextra -Werror`. Fails, and nothing after it runs. |
| Required constructs | the mechanism the item teaches is in the source; comments do not count |
| Public tests | the tests `make check` ran |
| Hidden tests | the same behaviour on inputs you have not seen: odd sizes, tiny sizes, awkward rank counts |
| Determinism | identical output over repeated runs at several thread counts |
| Speedup | the faster version is faster by a modest factor on a two-core machine; only checked once the code is correct |
| Analysis | your program computes speedup, efficiency, Karp-Flatt or roofline quantities exactly on hidden data |
| Committed results | the CSV you produced on the cluster has the right columns, rows and ranges |
| Answers | numbers in `answers.yaml` against a key, with a tolerance |

Some marks cannot be earned on the grading machine. Items marked *GPU cluster* in the tables have those checks run on `asaicompute` after the deadline and added to your total; the report tells you how many marks are pending. The four assignments carry 4 marks for a short written section, `EXPLANATION.md`, graded against the rubric printed in the README. Everything else is marked by the tests, and the last submission before the deadline is the one that counts.

# Part 3: The items

Accept links are per section. `A` is `aim-a-hpc`, `B` is `aim-b-hpc`. Due times are 23:59 IST on the day shown. Late submissions are marked out of 80 percent within 48 hours and not accepted after that, as in the handbook.

**Every item is open now.** The deadlines are spread evenly from mid-September to 23 October, the last working day of the semester, in syllabus order, so that one item falls due roughly every other day. Nothing stops you from working ahead: accept whatever you like now, and submit early. The order in the tables is the order the material is taught, and the sensible order to do them in.

Everything that needs the GPU cluster now closes on **Tuesday 20 October**: Labs 2, 3 and 8, Exercises 17 and 19, and Assignments 1, 2 and 3, alongside Exercise 20 which was already on that date. You have no accounts on `asaicompute` yet, and it would be unfair to hold you to a deadline for work you cannot start. Details of how to log in will follow separately; the rest of each of those items can be done now.

Lab 1 and Exercises 01 and 02 close on **Monday 28 September** as already announced.

> **PITFALL · ONE ITEM EVERY TWO DAYS IS THE PACE, NOT A SUGGESTION**
>
> The dates are close together on purpose. An item left for the weekend before the deadline collides with the two behind it. Do each one the week its material is covered and the schedule is light; leave them and it is not.

## Warm-up (no marks)

| ITEM | NAME | MARKS | ACCEPT |
|---|---|---|---|
| `a0-git-warmup` | A0: Git and GitHub warm-up | 0 | A · B |

Do A0 first. It is the whole loop above on a task that cannot hurt you, and it tells us both that your account, your tools and the grader agree.

## Coding exercises (2% of the course, pooled)

| ITEM | NAME | MARKS | DUE (IST) | ACCEPT | GRADED WHERE |
|---|---|---|---|---|---|
| `ex01-performance-metrics` | Ex 01: Performance metrics and the roofline | 10 | Mon 28 Sep, 23:59 | A · B | runner only |
| `ex02-cache-blocking` | Ex 02: Cache blocking | 12 | Mon 28 Sep, 23:59 | A · B | runner only |
| `ex06-openmp-reduction` | Ex 06: Parallel reduction with OpenMP | 10 | Mon 28 Sep, 23:59 | A · B | runner only |
| `ex07-openmp-load-balance` | Ex 07: Load imbalance and the schedule clause | 12 | Tue 29 Sep, 23:59 | A · B | runner only |
| `ex08-openmp-histogram` | Ex 08: Parallel histogram | 12 | Thu 01 Oct, 23:59 | A · B | runner only |
| `ex11-mpi-halo-stencil` | Ex 11: MPI halo exchange | 10 | Thu 08 Oct, 23:59 | A · B | runner only |
| `ex12-mpi-collectives` | Ex 12: MPI collectives | 11 | Sat 10 Oct, 23:59 | A · B | runner only |
| `ex17-cuda-vector-add` | Ex 17: First CUDA kernels | 10 | Tue 20 Oct, 23:59 | A · B | 7 on the GPU cluster |
| `ex19-cuda-tiled-matmul` | Ex 19: Shared-memory tiling | 12 | Tue 20 Oct, 23:59 | A · B | 7 on the GPU cluster |
| `ex20-cuda-reduction` | Ex 20: Parallel reduction on a GPU | 10 | Tue 20 Oct, 23:59 | A · B | 7 on the GPU cluster |

Each exercise is one file in `src/`. The starters compile and are correct serial code; the marks are for the parallel or optimised version the statement asks for, and for keeping it correct on the hidden inputs.

## Lab sheets (2% of the course, pooled)

| ITEM | NAME | MARKS | DUE (IST) | ACCEPT | GRADED WHERE |
|---|---|---|---|---|---|
| `lab01-timing-and-memory` | Lab 1: Timing and memory | 10 | Mon 28 Sep, 23:59 | A · B | runner only |
| `lab02-slurm-and-the-cluster` | Lab 2: SLURM and the cluster | 10 | Tue 20 Oct, 23:59 | A · B | runner only |
| `lab03-scaling-study-and-roofline` | Lab 3: Scaling study and roofline | 10 | Tue 20 Oct, 23:59 | A · B | runner only |
| `lab04-openmp-and-races` | Lab 4: OpenMP and races | 10 | Sat 26 Sep, 23:59 | A · B | runner only |
| `lab05-optimising-matrix-multiply` | Lab 5: Optimising matrix multiply | 10 | Sat 03 Oct, 23:59 | A · B | runner only |
| `lab06-mpi-basics-and-deadlock` | Lab 6: MPI basics and deadlock | 10 | Tue 06 Oct, 23:59 | A · B | runner only |
| `lab07-domain-decomposition` | Lab 7: Domain decomposition | 10 | Sun 11 Oct, 23:59 | A · B | runner only |
| `lab08-first-cuda-kernels` | Lab 8: First CUDA kernels | 10 | Tue 20 Oct, 23:59 | A · B | 3 on the GPU cluster |
| `lab09-cuda-profiling-and-roofline` | Lab 9: CUDA profiling and roofline | 10 | Wed 21 Oct, 23:59 | A · B | runner only |

The lab sheets keep their original teaching text; the last section of each README, *What is graded*, says which deliverables the tests look at. Measurements that need the cluster are committed as CSVs and shape-checked; the numbers themselves are yours to interpret.

## Assignments (6% of the course, pooled)

| ITEM | NAME | MARKS | DUE (IST) | ACCEPT | GRADED WHERE |
|---|---|---|---|---|---|
| `a1-measure-before-you-optimise` | Assignment 1: Measure before you optimise | 30 | Tue 20 Oct, 23:59 | A · B | 4 by rubric |
| `a2-openmp-scaling` | Assignment 2: OpenMP scaling | 30 | Tue 20 Oct, 23:59 | A · B | 4 by rubric |
| `a3-mpi-decomposition` | Assignment 3: MPI decomposition | 30 | Tue 20 Oct, 23:59 | A · B | 4 by rubric |
| `a4-gpu-and-ai` | Assignment 4: GPU and AI | 30 | Fri 23 Oct, 23:59 | A · B | 14 on the GPU cluster; 4 by rubric |

Thirty marks each: 26 from tests, 4 from `EXPLANATION.md` (at most 250 words, rubric in the README). The kernel is fixed for everyone so that the tests are the same for everyone: Jacobi stencil, CSR sparse matrix-vector product, 2-D heat diffusion with MPI, and a CUDA matrix transpose.

# Part 4: The GPU cluster

Eleven items need a GPU or a real multi-core machine: Labs 2, 3, 8 and 9, Exercises 17, 19 and 20, and all four assignments. You do not get your own cluster account; the class shares one, `hpccourse`, on `asaicompute`.

## Getting in

Access is by **your own GitHub SSH key**, the same one you use to push. Nothing to request and no password: if your key is on your GitHub account, it already works.

```
ssh hpccourse@172.17.16.11
```

You land in your own directory, `w/<your roll number>`, and the prompt tells you so. If you are refused, your GitHub account has no SSH key on it; add one at github.com/settings/keys and tell me so I can refresh the list.

**PITFALL · YOU MUST BE ON THE CAMPUS VPN, NOT JUST CAMPUS WIFI**

The student wifi range is blocked from reaching the servers. Connect to the Amrita VPN first, from anywhere including on campus, and `172.17.16.11` becomes reachable. Students who have got in so far all came through the VPN.

## Copying work across

`rsync` and `scp` both work. Clone your repository on your laptop as usual, then push the files up:

```
rsync -av ./src/ hpccourse@172.17.16.11:src/
scp results.csv hpccourse@172.17.16.11:
```

Pull results back the same way, and commit them from your laptop. Do not try to run `git` on the cluster; the account is deliberately limited to building and running.

## Running something

Two commands, and only these two. They set the partition, the account and the limits for you:

```
course-run make check            # run it now, on a GPU
course-submit scaling.sbatch     # queue a batch script
course-status                    # see the queue
```

Both cap a job at **five minutes** and one GPU. That is ample: these assignments run in one or two minutes. Nothing else needs a flag.

> **PITFALL · JOBS ONLY RUN BETWEEN 17:00 AND 07:00**
>
> The cluster belongs to researchers during the working day, so student jobs are confined to the evening and night. You may submit at any time: a job sent at noon sits as `PartitionDown` or "reservation not usable now" and starts by itself at 17:00. Plan for that the night before a deadline, not the afternoon of it.

Check on your jobs with **`course-status`**, and stop one with `scancel <jobid>`. Use `course-status` rather than plain `squeue`: the course partition is hidden from the cluster-wide view, so `squeue` on its own will not list these jobs. Everybody's jobs appear there; yours carry your roll number as the job name.

A job waiting with reason `(Reservation)` or `(PartitionDown)` is simply outside the window. It will start at 17:00 without you doing anything.

## Share the space sensibly

It is one account, so **everyone can see everyone's directories**. Keep nothing private there, and copy nothing out of anybody else's. Submissions are compared for similarity, and marks come from what you push to your own repository, so copying buys you nothing and costs you a great deal.

# Part 4: When something goes wrong

| SYMPTOM | WHAT IT MEANS |
|---|---|
| `gh student accept` says the assignment is not found | You are in the other section, or the item is not released yet. |
| `git clone https://...` asks for a password | Use `gh repo clone` or the `git@github.com:` address. |
| `make check` passes, the report shows hidden tests failing | Your code is right for the sizes you tried. Read the failing test's name: it names the case. |
| `! [rejected] main -> main (fetch first)` on push | Submitting added a commit. `git pull --rebase origin main`, then push again. |
| Speedup check says *skipped: ... failed* | Speed is only measured once correctness passes. Fix the tests first. |
| The report says marks are graded on the cluster | Normal for CUDA items. They arrive after the deadline. |
| No release two minutes after submitting | Open the Actions tab. If the run itself is red, send me the link. |
| `nvcc: command not found` on your laptop | CUDA items build on the cluster (Part 4), not on a laptop without a GPU. |
| `ssh hpccourse@...` refused | Your GitHub account has no SSH key, or you are off the campus network. |
| Job stuck at `PartitionDown` | Normal outside 17:00 to 07:00. It starts on its own at 17:00. |
| `Requested reservation not usable now` | Same thing: you are outside the window. |